INCH-POUND

MIL-PRF-49150D(CR)
4 March 1999
SUPERSEDING
MIL-E-49150C(CR)
26 May 1992

PERFORMANCE SPECIFICATION
ENTRANCE HOUSING ASSEMBLY
VIEWER, DRIVER'S, NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Entrance Housing Assembly which is part of Viewer, Driver's Night Vision, AN/VVS-2(V).

1.2 Classification. Entrance housing assemblies covered by this specification must be the following types as specified;

Type I	Entrance Window and Mirror Housing Assembly for AN/VVS-2(V)1A (M60 Application)
Type II	Prism Objective Housing Assembly for AN/VVS-2(V)2A (XM-1 Application)
Type III	Prism Objective Housing Assembly, for AN/VVS-2(V)3 (FVS Application)
Type IV	Prism Objective Housing Assembly for AN/VVS-2(V)4 (Paladin Application)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document by letter.

AMSC N/A FSC 5855

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2.1.1 Reference documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS
USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-77133/SM-D-771476	- Entrance Housing Assembly (V1A Configuration)
SM-D-77143/SM-D-771486	- Prism Objective Housing Assembly (V2A/V3 Configuration)
5005814	- Prism Objective Housing Assembly (V4 Configuration)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2. Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The entrance housing assembly is herein referred to as the entrance assembly. The entrance assembly is the portion of the driver's viewer outside the vehicle. It directs the line of sight to the objective lens.

3.2 Construction. The entrance assembly shall be constructed to assure compliance with the contract configuration baseline and in accordance with the following drawings as applicable (see 6.2.d):

Type I	SM-D-771333/SM-D-771476	Entrance Housing Assembly
Type II/III	SM-D-771434/SM-D-771486	Prism Objective Housing Assembly
Type IV	5005814	Prism Objective Housing Assembly

3.3 First article testing (FAT). When specified in the contract (see 6.2.c and 6.2.g) the contractor shall furnish unit(s) for FAT inspection and approval (see 4.2).

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3.4 Materials. The contractor shall select parts and materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.5 Performance characteristics.

3.5.1 Angular deflection. An optical ray perpendicular to the mounting flange shall be deflected through an angle of 108, ± 2 degrees for Type I and 90, ± 2 degrees for Type II, Type III and Type IV.

3.5.2 Resolution. The resolution shall be 7 line-pairs/milliradian or better when rotated ±15 degrees about a center 35mm from the mounting flange and a telescope of 8 power (X) or greater with a 30mm aperture is used. The telescope may be focused a maximum of 0.03 diopter from infinity for best focus.

3.5.3 Transmission. The transmission shall be 80 percent minimum over the entire aperture in the spectral region defined by a 2854K, ± 200K source and an S-20 extended-red (ER) detector.

3.6 Environmental.

3.6.1 Temperature Extremes. The entrance assembly shall not be damaged nor suffer degradation of performance (3.10.1 and 3.10.4) after being subjected to storage at +68°C and -51°C. The entrance assembly shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to repetitive temperature changes between +23°C and -51°C within 5 minutes and between +23°C and +68°C within 5 minutes. The soak time at each temperature extreme shall be sufficient for the entire entrance assembly to achieve stabilization at the specified temperature. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature.

3.6.2 Humidity. The entrance assembly shall not be damaged nor suffer degradation of performance after being subjected to +68°C and 94 ±4 percent humidity (see 3.10.1 and 3.10.4).

3.6.3 Immersion. The entrance assembly shall not be damaged (see 3.10.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The entrance assembly shall be pre-conditioned by exposure to temperature of +45°C ±3°C for a period of 2 hours prior to being immersed in the fresh water at a temperature of +18°C ±5°C.

3.6.4 Vibration. The entrance assembly shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation. Vibration shall be limited to one sine sweep from 5.0 to 500 Hertz (Hz.) and back to 5.0 Hz

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within 15 minutes in the vertical axis only. Vibration shall be performed from 5.0 to 5.5 Hz. at 1 inch displacement, 5.5 to 30 Hz. at 1.5 g’s peak, 30 to 50 Hz. at 0.33 inch displacement and 50 to 500 Hz. at 4.2 g’s peak.

3.6.5 Shock.

3.6.5.1 Basic shock. The entrance assembly shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to basic shock of 40g's (see 3.10.5) with a duration of 18 milliseconds.

3.6.5.2 Weapon shock. The entrance assembly shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to simulated weapon firing shock of 100g's (see 3.12.5) with a duration of 1.5 milliseconds one half sine pulse.

3.7 Treatment and painting. Treatment and painting shall be anti-reflective, shall be rust resistant and as specified on applicable drawings unless otherwise specified in the contract.

3.8 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified in the contract.

3.9 Workmanship. Standards of workmanship shall be such that the entrance assembly will meet all requirements of this specification and any referenced specifications or drawings.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3 or 4, mandatory for this specification.

3.10.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.10.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.10.3 Environmental temperature. Environmental chamber temperatures shall be controlled to within ± 5°C of the specified temperature unless otherwise specified.

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3.10.4 Degradation of performance. Degradation of performance is defined as failure to operate in accordance with the performance requirements of this specification. For post-environmental test verification, degradation of performance shall be defined as failure to meet the performance requirements of this specification.

3.10.5 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be classified as follows:

a. First article testing (4.2).
b. Conformance inspection (4.3)

4.2 First article testing. When specified in the contract or purchase order, first article testing shall be performed by the contractor. The quantity of first article test samples shall be as specified in the contract or purchase order (see 6.2.c) and shall be selected from the first production lot.

4.2.1 FAT testing. Each entrance assembly in the first production lot shall be subjected to the tests of Table I. After completion of this testing, entrance assemblies selected as samples for FAT testing shall be subjected to the tests of Table II which may be performed in any order with the exception that immersion is last. Failure of any test shall be cause for rejection of that entrance assembly and may be cause for rejection of the FAT.

TABLE I. Unit first article testing.

Inspection	Requirement Paragraph
Angular deflection	3.5.1
Resolution	3.5.2
Transmission	3.5.3

Table II Sample first article testing.

Inspection	Requirement Paragraph
Temperature Extremes	3.6.1
Humidity	3.6.2
Immersion	3.6.3
Vibration	3.6.4
Shock	3.6.5

4.2.2.1 Disposition of FAT samples. Disposition of FAT samples shall be as specified in the contract or purchase order.

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.f). Unless otherwise specified all tests shall be performed at ambient temperature (see 3.12.2).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirement shall be as specified in the contract or purchase order (see 6.2.h). When actual packaging is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The entrance assembly is used with the AN/VVS-2(V) for driver's viewing during periods of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Necessary actions by contractor in the event of a lot failure.
c. Quantity and schedule for First Article testing for all line items and disposition of FAT samples.
d. Type of entrance assembly (see 3.2).
e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
f. Statement identifying that the contractor must submit his proposal for conformance inspection to include specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
g. Statement specifying that FAT testing is, or is not, required.
h. Packaging requirements.
i. Statement specifying that Coated Optical Component coating is, or is not, required.

6.3 Definitions. See 3.10.

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6.4 Subject term (key word) listing.

Angular deflection
Resolution
Transmission

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodian:
Army - CR

Preparing Activity:
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Project 5855-0093

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49150D (CR)	**2. DOCUMENT DATE (YYMMDD)** 990304

3. DOCUMENT TITLE
Entrance Housing Assembly Viewer, Driver's, Night Vision, AN/VVS-2(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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